

Assignment - 3

VM Image Creation, Deployment & Replication

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Objective

To create a custom Virtual Machine image, launch a new VM instance from that image, and replicate it across Azure regions.

Task 1: Create a VM with httpd Service (User Data Script)

Steps to provision the initial virtual machine with auto-configured Apache web server:

1. Navigate to Azure Portal and start the **Create Virtual Machine** wizard.
2. Under the **Advanced** tab, enable the **User Data** checkbox.
3. Enter the following script to automate Apache configuration during installation:

```
#!/bin/bash
sudo apt-get update -y
sudo apt-get install apache2 -y
sudo systemctl start apache2
sudo systemctl enable apache2
echo "<h1>Azure Web Server - Apache Deployed Successfully via User Data</h1>" | sudo tee /
var/www/html/index.html
```

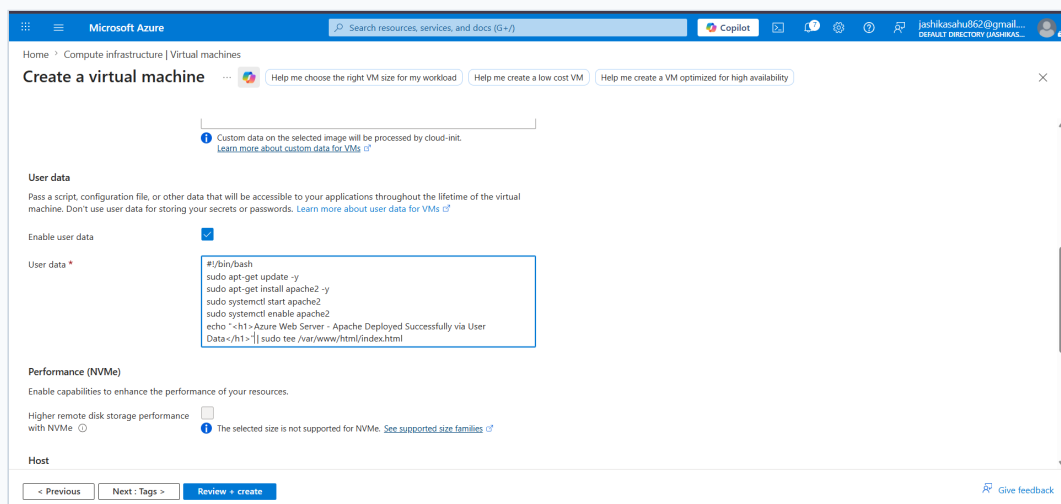


Fig 1.1: Virtual machine creation wizard with User Data script setup.

Verify that the web server is live by copying the Public IP address and pasting it into a web browser tab.

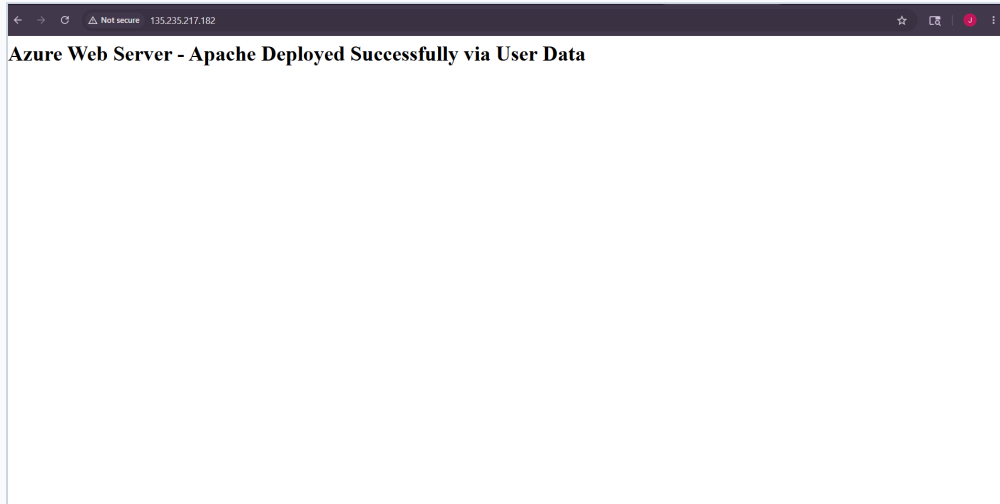


Fig 1.2: Apache web server index page live at Public IP 135.235.217.182.

Task 2: Create a Specialized Image from the VM

Steps to stop the Virtual Machine instance and capture a specialized image for environment replication:

1. Open the virtual machine dashboard and click **Stop** to deallocate resources.
2. Click **Capture** on the upper action bar to generate a reusable platform template.
3. Set the image state as **Specialized** to preserve configuration properties.
4. Submit the gallery info and hit **Create** to trigger the cloud image pipeline.

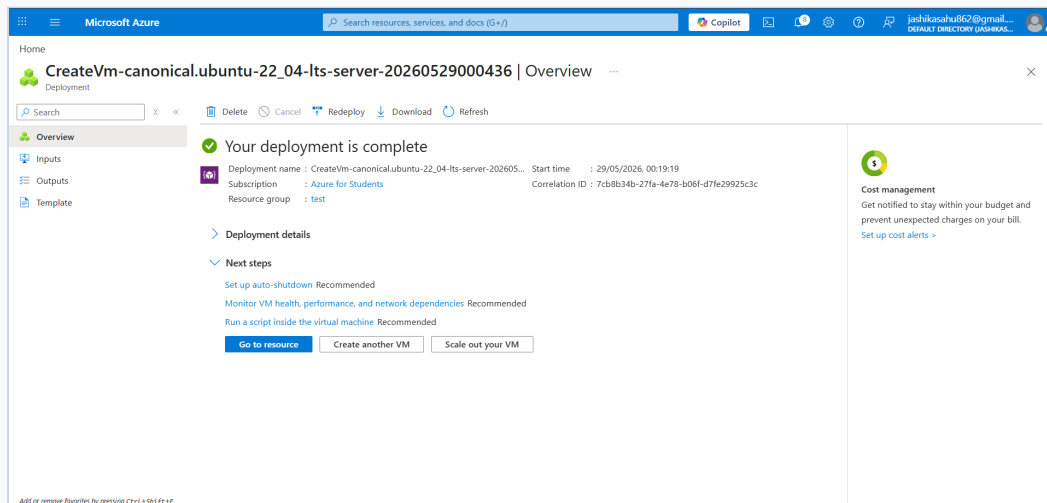


Fig 2.1: Deployment complete screen for the virtual machine image capture sequence.

Task 3: Launch a New VM Using the Captured Image

Steps to deploy a clone system node utilizing the managed custom gallery image:

1. Navigate into the Azure Compute Gallery workspace asset and locate the custom image definition.
2. Click **+ Create VM** directly from the image version dashboard.
3. Complete standard network bindings and trigger the structural deployment template wizard.

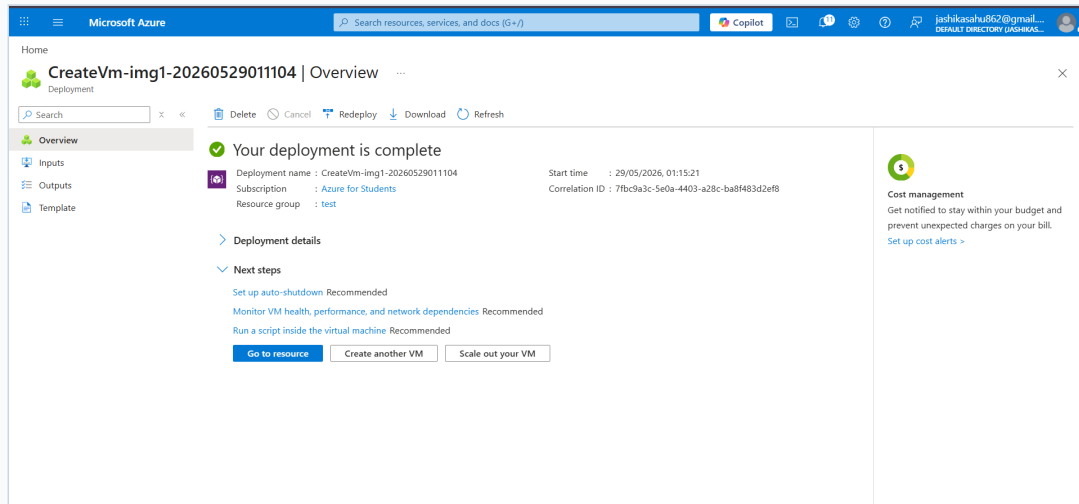


Fig 3.1: Azure Portal deployment completion overview page for the custom image template instance.

Task 4: Check Output - Verify httpd is Running

Steps to test the functionality of the newly cloned virtual machine instance:

1. Open the resource panel of the new virtual machine to extract its assigned network gateway properties.
2. Copy the new **Public IP Address** (40.81.232.61).
3. Open a web browser tab and navigate to the IP address to verify the active web server.

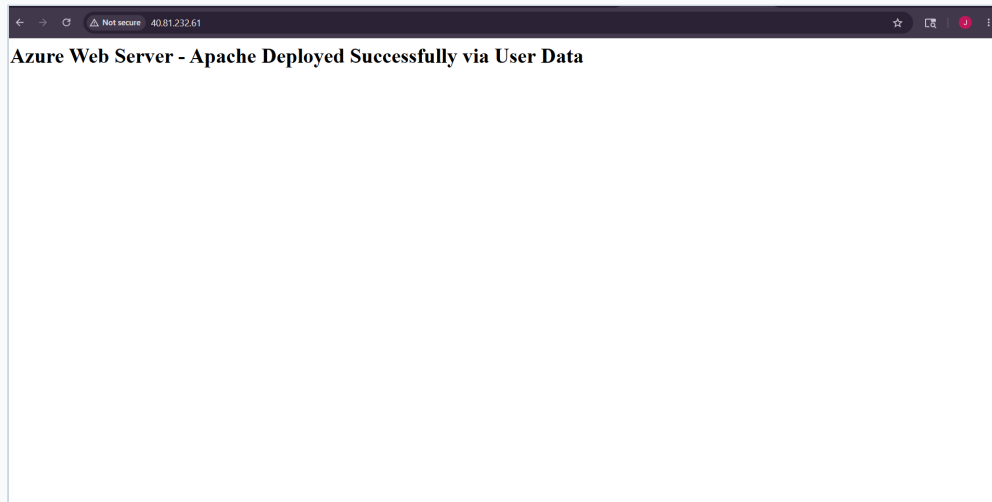


Fig 4.1: Browser showing the successful response page live from Public IP 40.81.232.61.

Task 5: Replicate Image to Malaysia East & Verify

Steps to distribute the custom image asset to target cloud regions for regional validation:

1. Open the image version settings panel and navigate to the **Update replication** tab section.
2. Add **Malaysia East** as the target cloud geographical location region.
3. Click **Save** to execute sync processes across datacenters.

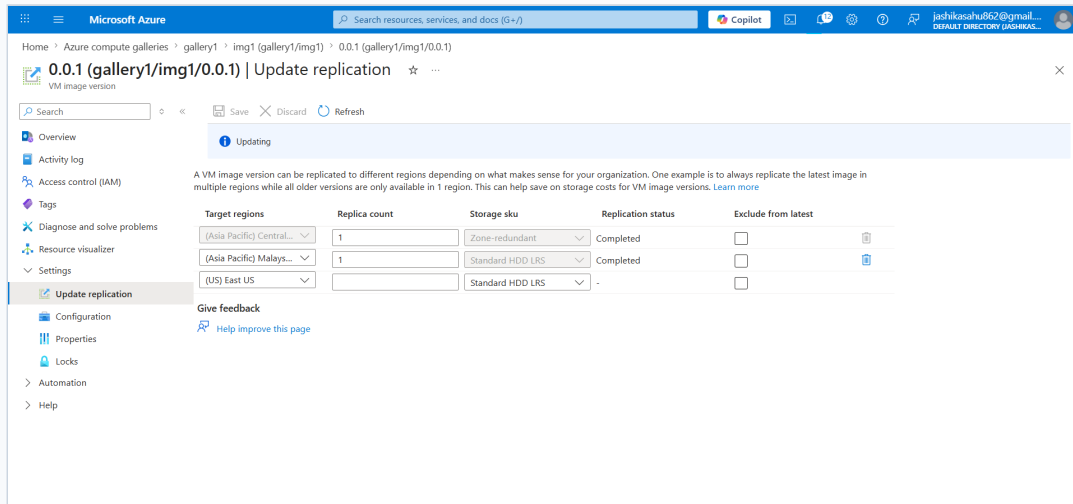


Fig 5.1: Replication panel status displaying target region updates for Malaysia East distribution.

Launch an isolated test machine in the updated regional slot zone and access its endpoint URL to verify integrity.

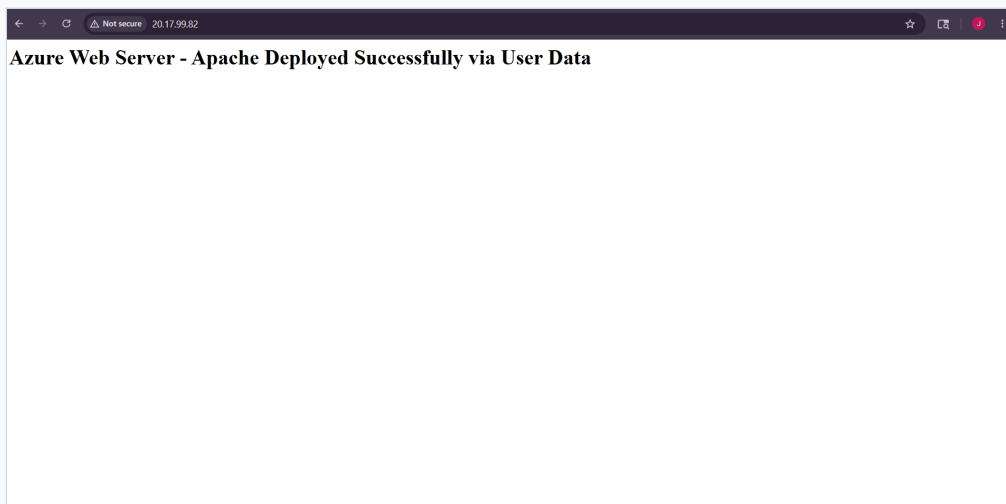


Fig 5.2: Browser view loading the replicated Apache web server response from Public IP 20.17.99.82.